

Introduction

The heart is not a metronome. The interval between each heartbeat is constantly varying in response to input from the brain. This variability is considered to be an index of the brain's ability to adaptively respond to ever-changing internal and external input. There are various ways to quantify this heart rate variability (HRV) but, importantly, lower variability has been tied to poorer physical health (e.g., inflammation, cardiovascular disease risk and mortality) and poorer mental health (e.g., major depression).¹⁻³ Although the impact of chronic stress on physical health is well established, findings on the association between stress and HRV are mixed, potentially due to the variety of measures for HRV and stress used.⁴⁻⁵ To further explore this association, we examined the relationship of vagally-mediated HRV with two self-report stress measures: self-perceived stress and past-month stressful life events.

REFERENCES:

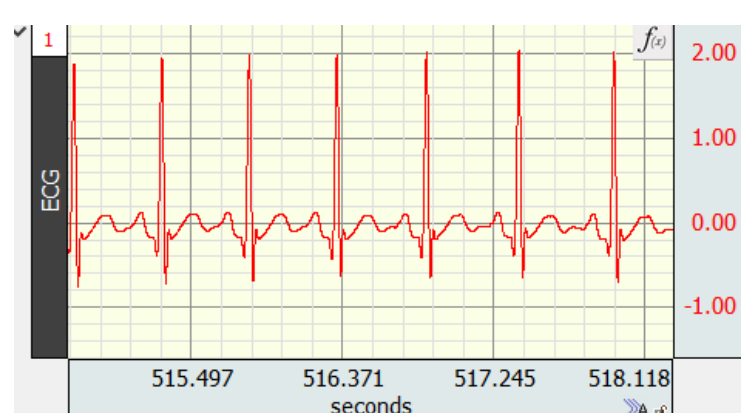
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Methods



Participants were forty female undergraduates

- Age: 18-24yrs (M = 19.1 ; SD = 1.6)
- 95% identified as Hispanic (of whom 50% identified as Caucasian, 50% identified as non-Caucasian)

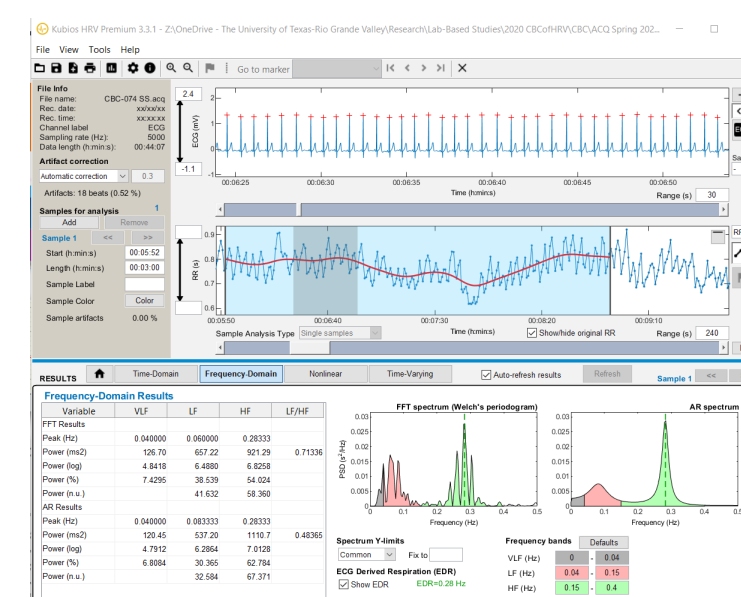


ECG recorded on a Biopac system using a Lead II configuration

- A 3-minute recording was obtained while participants sat quietly, breathing spontaneously

Participants then completed an online survey which contained two standardized measures of stress:

- **Perceived Stress Scale (PSS).** A 10-item measure to assess the level of self-perceived stress. Questions include, "In the last month, how often have you found that you could not cope with all the things that you had to do?" Possible scores range from 0 – 40, with higher scores indicating greater stress.
- **Inventory of College Students' Recent Life Experiences (ICSRL).** The ICSRL is comprised of 49-items that reflect everyday stressors or hassles commonly experienced by college students. For each item, respondents rated the intensity of their experience with the stressor over the past month on a four-point Likert scale (1: not at all part of my life to 4: very much part of my life). Possible scores range from 49 to 196 and are calculated by summing up the 49 item ratings. Higher ICSRL scores represent more stress.



Kubios HRV analysis software version 3.3.1 was used to visually inspect the ECG recording for signal quality and artifact.

- Automatic artifact corrections were applied when artifact was below 5%.
- To assess vagal activity indexed by heart function, RMSSD (square root of the mean squared differences of successive R-R intervals) and HfHRV (high frequency .15–.4 Hz, based on spectral analysis) values were calculated.
- Data from two participants were excluded due to poor/missing ECG. HRV values were log-transformed.

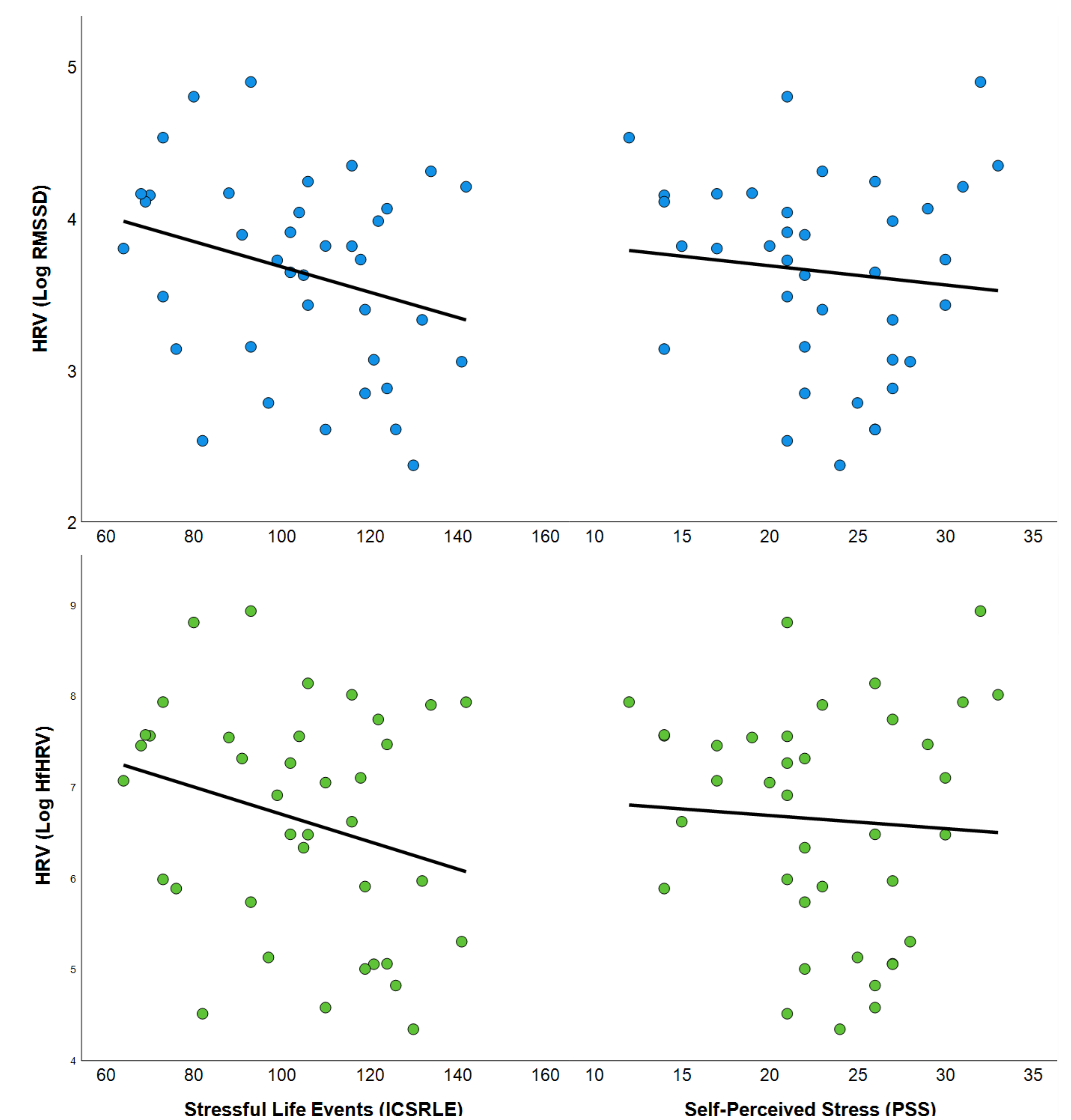
Conclusion

Our findings suggest that the association between vagally-mediated HRV and self-reported measures of chronic stress may depend on the manner by which stress is assessed. Such knowledge may be important in the development of future research, either as a variable to be controlled for or a variable of interest. The study was based on a primarily Hispanic sample of young adult female college students which limits the generalizability of findings. Additionally, given the small sample size in this study, it is possible that the observed associations (and lack of association) are spurious. A clearer understanding of this issue will require additional data in a more heterogenous sample.

Results

The two stress measures were significantly positively correlated ($r(36) = .68, p < .001$). The two HRV measures were significantly positively correlated ($r(36) = .99, p < .001$). Age was significantly negatively correlated with both RMSSD and HfHRV ($-.37$ and $-.36$, respectively). Partial correlations, controlling for age, revealed a **significant negative correlation between HRV and life-events stress** (RMSSD, $r(35) = -.35, p = .016$; HfHRV $r(35) = -.33, p = .025$), but not self-perceived stress (RMSSD, $r(35) = -.14, p > .05$; HfHRV $r(35) = -.09, p > .05$).

Scatterplots of HRV measures against self-perceived stress and stressful life events



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